

Virtual Data Science with Python Apprentice Intern

Week 2 Report

Topic:

Advanced Data Visualization and Storytelling with Python

Dataset: Titanic Dataset (Kaggle)

Submitted By: Khalik Ansari

Internship: Virtual Data Science with Python Apprentice Intern

Week: 2

Submission Date: 04/08/2026

1. Objective

The objective of this task is to create meaningful and advanced data visualizations using Python and present them in the form of a data story. The analysis aims to communicate important insights from the dataset through clear and informative charts that can be easily understood by both technical and non-technical audiences.

2. Dataset Information

- **Dataset Name:** Titanic Dataset
- **Source:** Kaggle
- **Domain:** Transportation / Passenger Survival Analysis
- **File Format:** CSV
- **Total Rows:** 891
- **Total Columns:** 12

3. Methodology

The dataset was loaded into Python using the Pandas library. Missing values were handled by replacing the Age column with its median value and the Embarked column with its mode. The Cabin column was removed because it contained a large number of missing values. After preprocessing, various visualizations were created using Matplotlib and Seaborn to identify trends, compare different passenger groups, and communicate insights through data storytelling.

4. Libraries Used

The following Python libraries were used in this project:

- **Pandas** – For loading and preprocessing the dataset.
- **Matplotlib** – For creating charts and visualizations.
- **Seaborn** – For generating attractive statistical visualizations.

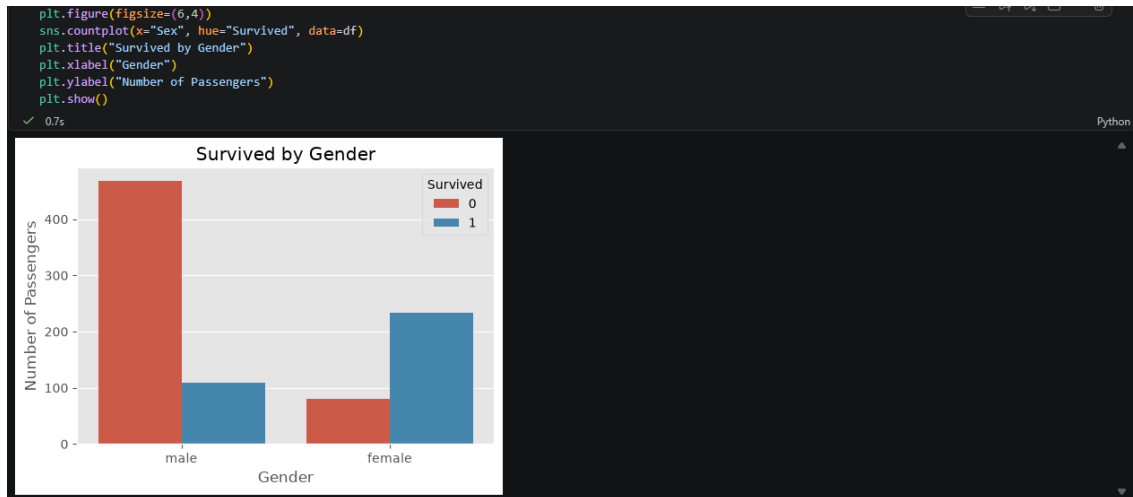
5. Visualization 1: Survival by Gender

Objective

This visualization compares the survival status of male and female passengers to understand how gender influenced survival during the Titanic disaster.

Code:

```
plt.figure(figsize=(6,4))
sns.countplot(x="Sex", hue="Survived", data=df)
plt.title("Survival by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")
plt.show()
```



Explanation

The count plot displays the number of male and female passengers who survived and who did not survive. Different colors represent the survival status, making it easy to compare both groups.

Data Story (Insight)

The visualization shows that **female passengers had a much higher survival rate than male passengers**. Although there were more male passengers on board, a large number of them did not survive. This suggests that gender played an important role in survival during the Titanic disaster and reflects the historical evacuation practice of giving priority to women.

6. Visualization 2: Survival by Passenger Class

Objective

The purpose of this visualization is to compare the survival status of passengers across different passenger classes (First, Second, and Third Class).

Code

```
plt.figure(figsize=(6,4))
sns.countplot(x="Pclass", hue="Survived", data=df)
plt.title("Survival by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")
plt.show()
```



Explanation

This count plot compares the number of passengers who survived and those who did not survive in each passenger class. It helps identify whether passenger class had an impact on survival.

Data Story (Insight)

The visualization shows that **First Class passengers had a higher survival rate than passengers in Second and Third Class**. In contrast, **Third Class had the highest number of deaths**, indicating that passengers traveling in lower classes faced greater difficulty during the disaster. This suggests that passenger class was an important factor influencing survival.

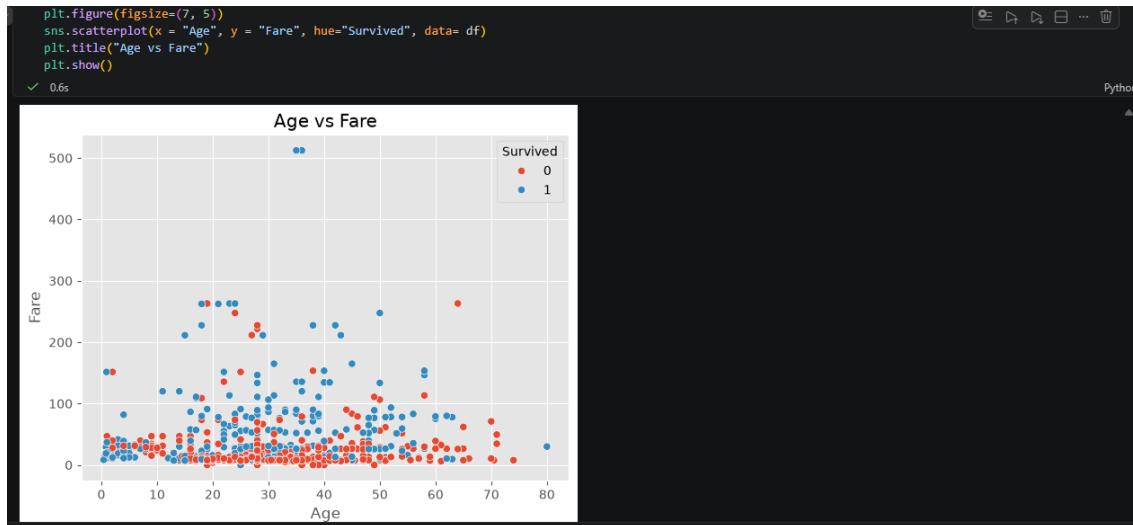
7. Visualization 3: Age vs Fare (Scatter Plot)

Objective

The objective of this visualization is to examine the relationship between passengers' age and the fare they paid. It also highlights the survival status of passengers to identify any noticeable patterns.

Code

```
plt.figure(figsize=(7, 5))
sns.scatterplot(x="Age", y="Fare", hue="Survived", data=df)
plt.title("Age vs Fare")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()
```



Explanation

The scatter plot displays the relationship between **Age** and **Fare**. Each point represents one passenger, while different colors indicate whether the passenger survived or not. This visualization helps identify trends, clusters, and possible outliers in the data.

Data Story (Insight)

The visualization shows that passengers who paid **higher fares** were generally concentrated in the upper fare range, while most passengers paid **lower fares**. There is **no strong relationship between age and fare**, as passengers of different ages paid a wide range of ticket prices. The plot also suggests that several passengers who paid higher fares survived, indicating that fare and passenger class may have influenced survival.

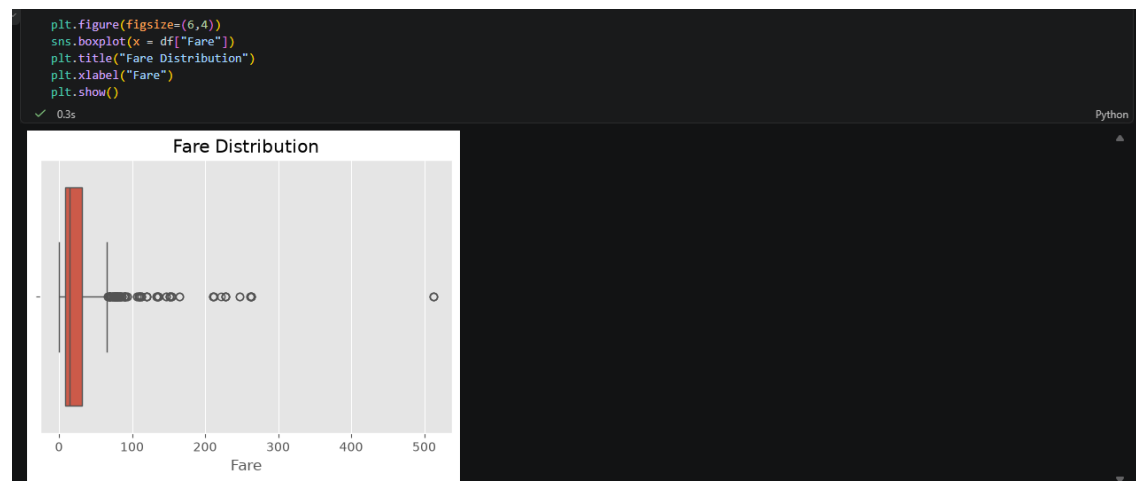
8. Visualization 4: Fare Distribution (Box Plot)

Objective

The objective of this visualization is to analyze the distribution of ticket fares and identify the presence of outliers in the dataset.

Code

```
plt.figure(figsize=(6, 4))
sns.boxplot(x=df["Fare"])
plt.title("Fare Distribution")
plt.xlabel("Fare")
plt.show()
```



Explanation

The box plot summarizes the distribution of ticket fares by showing the median, quartiles, and potential outliers. It is useful for understanding the spread of the fare values and detecting unusually high or low ticket prices.

Data Story (Insight)

The visualization shows that **most passengers paid relatively low ticket fares**, while a small number of passengers paid **significantly higher fares**. These high-fare values appear as **outliers** in the box plot. This suggests that although the majority of passengers traveled with affordable tickets, a few passengers—likely those in higher classes—paid much more for their journey.

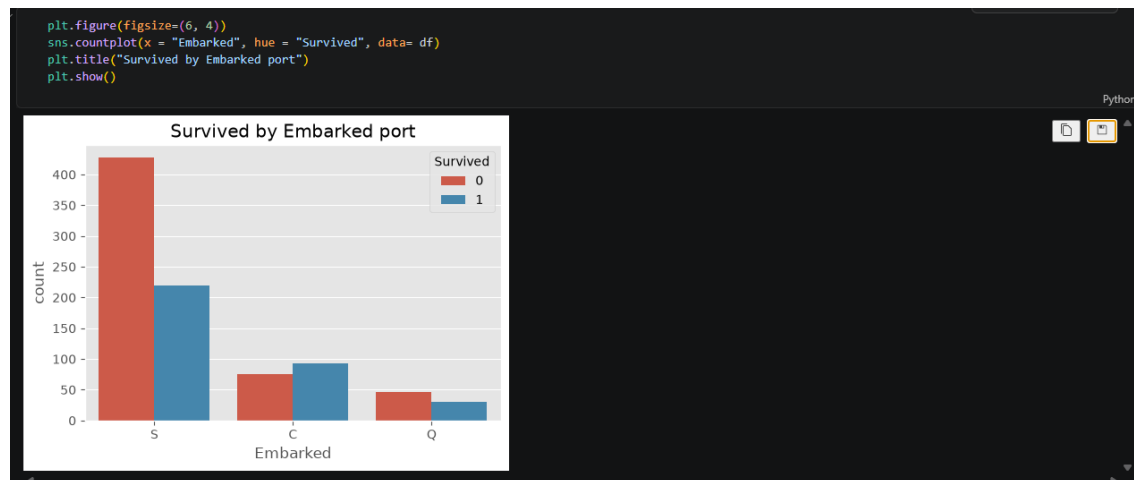
9. Visualization 5: Embarked vs Survival

Objective

The objective of this visualization is to compare the survival status of passengers based on their boarding port (**Embarked**). It helps identify whether the port of embarkation had any influence on passenger survival.

Code

```
plt.figure(figsize=(6, 4))  
sns.countplot(x="Embarked", hue="Survived", data=df)  
plt.title("Survival by Embarked Port")  
plt.xlabel("Embarked Port")  
plt.ylabel("Number of Passengers")  
plt.show()
```



Explanation

This count plot compares the number of passengers who survived and those who did not survive from each embarkation port (**C**, **Q**, and **S**). Different colors represent the survival status, making it easier to compare the outcomes across the three ports.

Data Story (Insight)

The visualization shows that **Southampton (S)** had the largest number of passengers, and consequently the highest number of survivors as well as non-survivors. **Cherbourg (C)** appears to have a relatively higher proportion of survivors, while **Queenstown (Q)** had the fewest passengers overall. This suggests that the boarding port may have had an indirect relationship with survival, likely because passengers from different ports belonged to different passenger classes and fare groups.

10. Business Implications

The visualizations created from the Titanic dataset provide meaningful insights that can support data-driven decision-making. Understanding the relationship between passenger class, gender, fare, and survival demonstrates how data visualization can reveal hidden patterns. Similar analytical techniques are widely used in industries such as transportation, healthcare, finance, and marketing to improve customer experience, assess risks, and support strategic planning. Presenting data through effective visualizations also helps both technical and non-technical audiences understand complex information more easily.

11. Key Insights

- Female passengers had a higher survival rate than male passengers.
- First Class passengers were more likely to survive than those in Second and Third Class.
- Most passengers paid lower ticket fares, while only a few paid very high fares.
- There was no strong relationship between passengers' age and ticket fare.
- Southampton (S) had the highest number of passengers, whereas Queenstown (Q) had the lowest.
- Data visualization made it easier to identify patterns and compare different passenger groups.

12. Conclusion

This project successfully demonstrated the use of advanced data visualization and storytelling techniques using Python. Different charts were created using Matplotlib and Seaborn to analyze the Titanic dataset from multiple perspectives. Each visualization highlighted a different aspect of the data and helped communicate

meaningful insights in a clear and understandable manner. The project improved my practical skills in creating effective visualizations, interpreting data, and presenting analytical findings through data storytelling.

13. References

- Kaggle – Titanic Dataset
- Pandas Documentation
- Matplotlib Documentation
- Seaborn Documentation